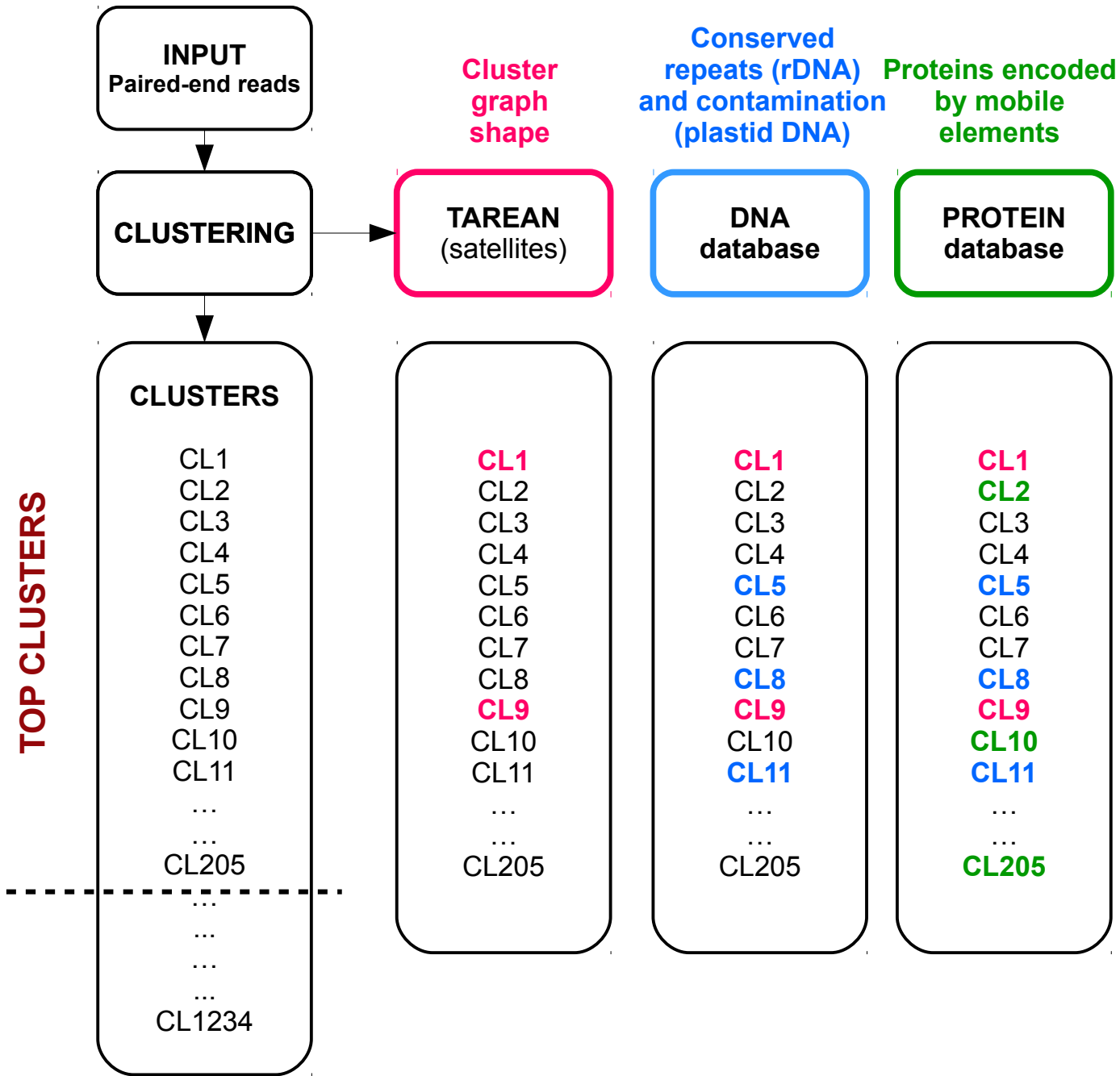
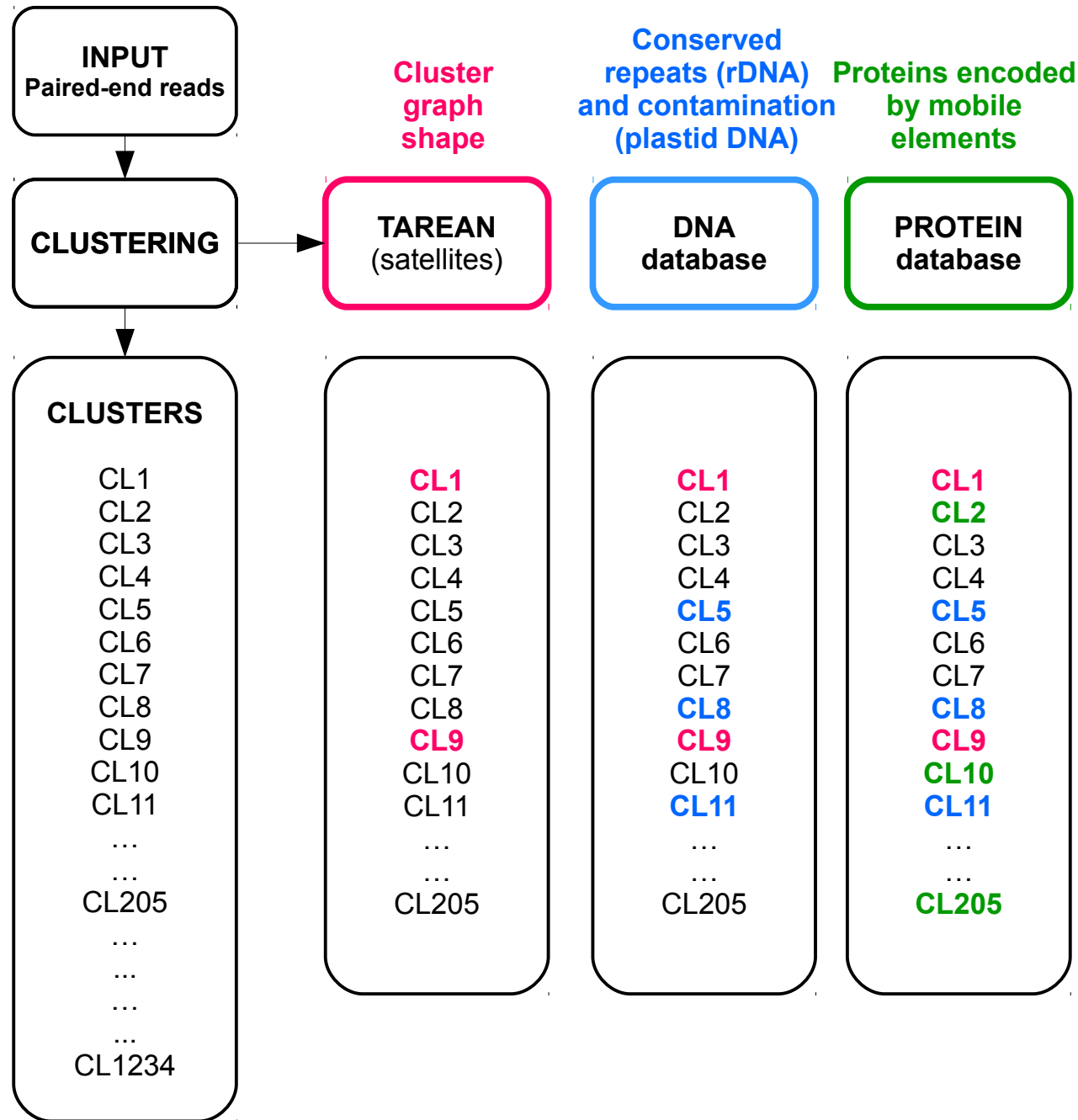


Using *RepeatExplorer* output for repeat
annotation and quantification

RepeatExplorer ver. 2



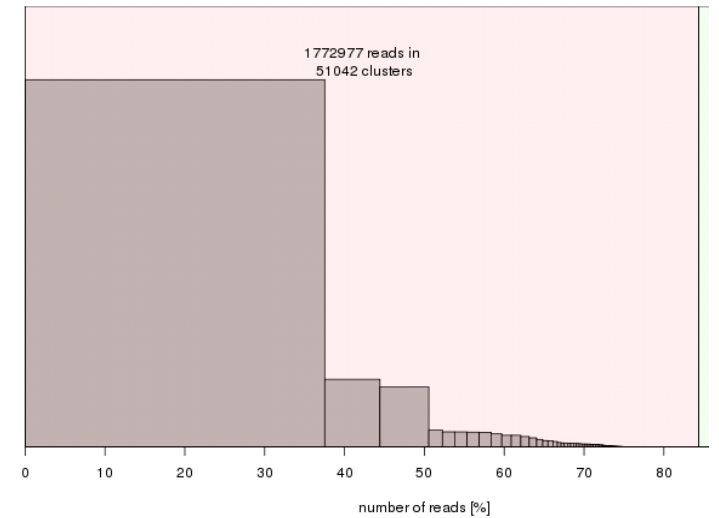
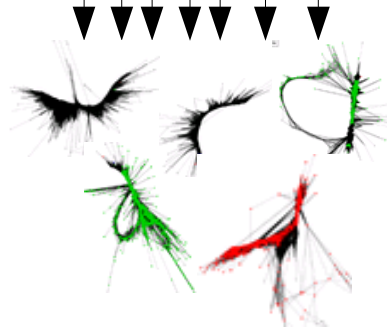
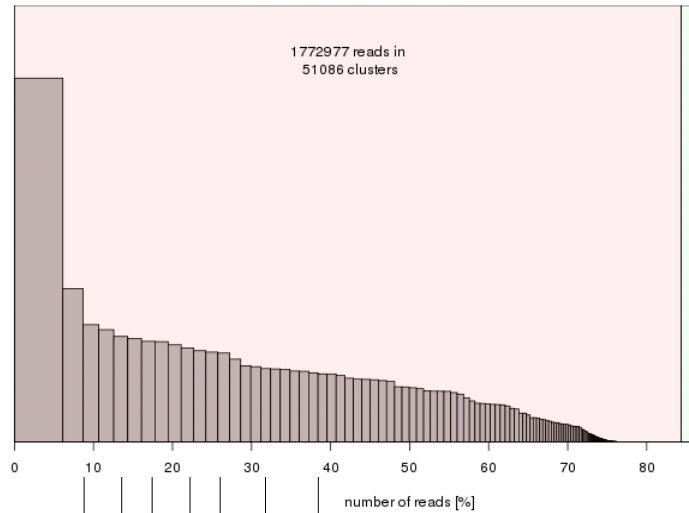
RepeatExplorer ver. 2



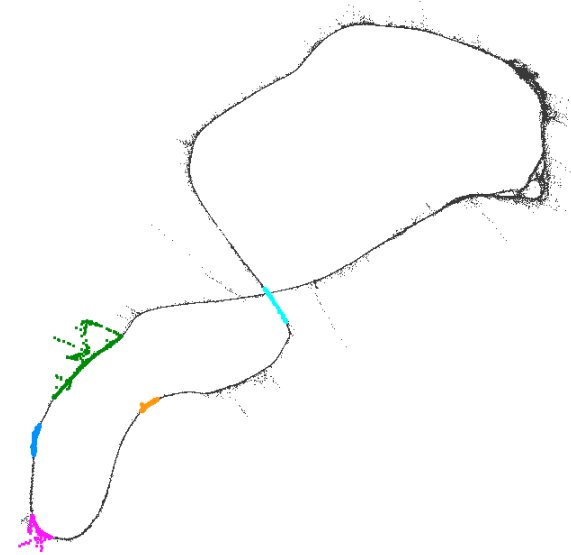
Many repeats are split
to multiple clusters

(feature of the clustering
algorithm; depends on
many factors)

Cluster fragmentation

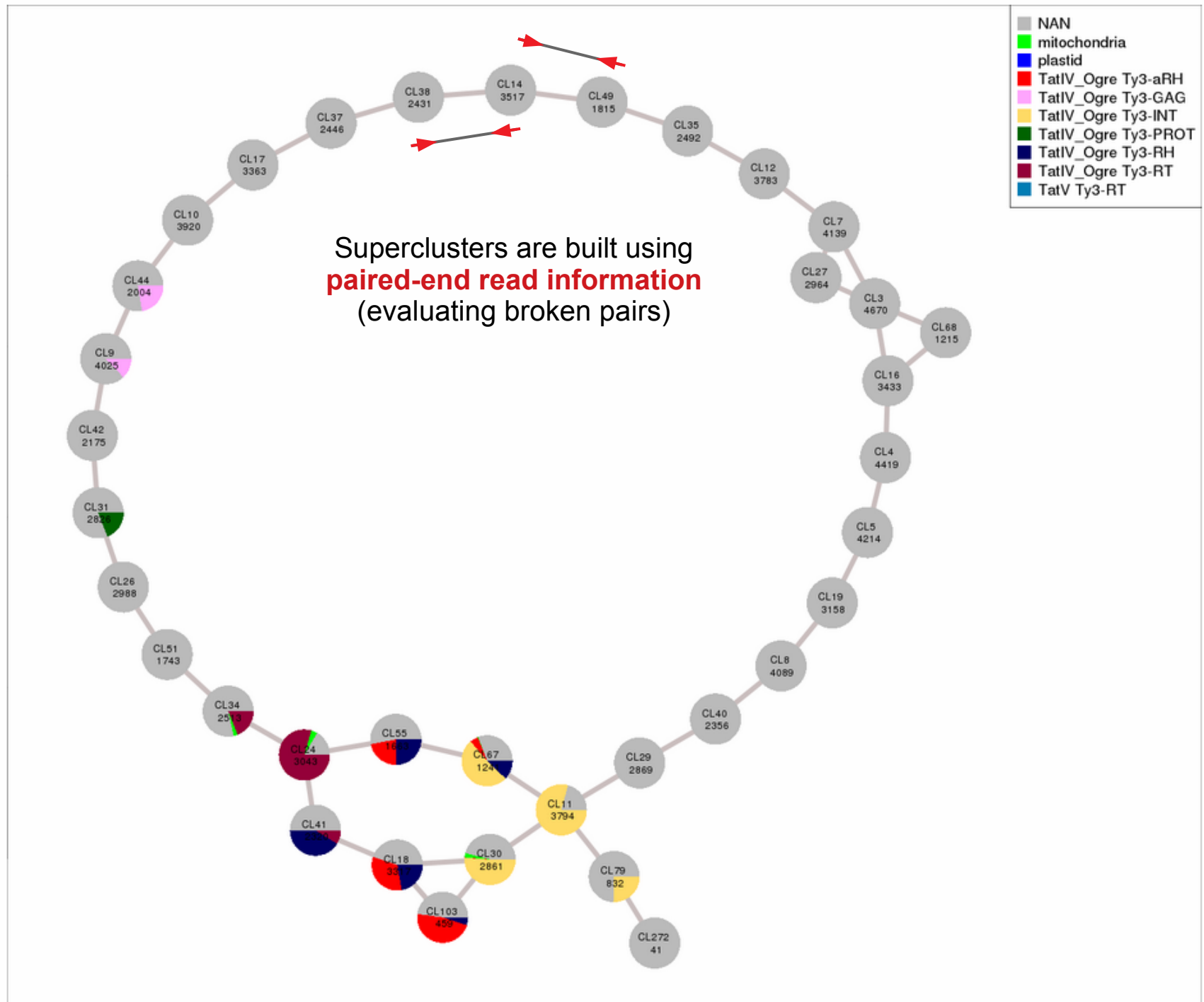


Evaluation of cluster connections via paired-end reads

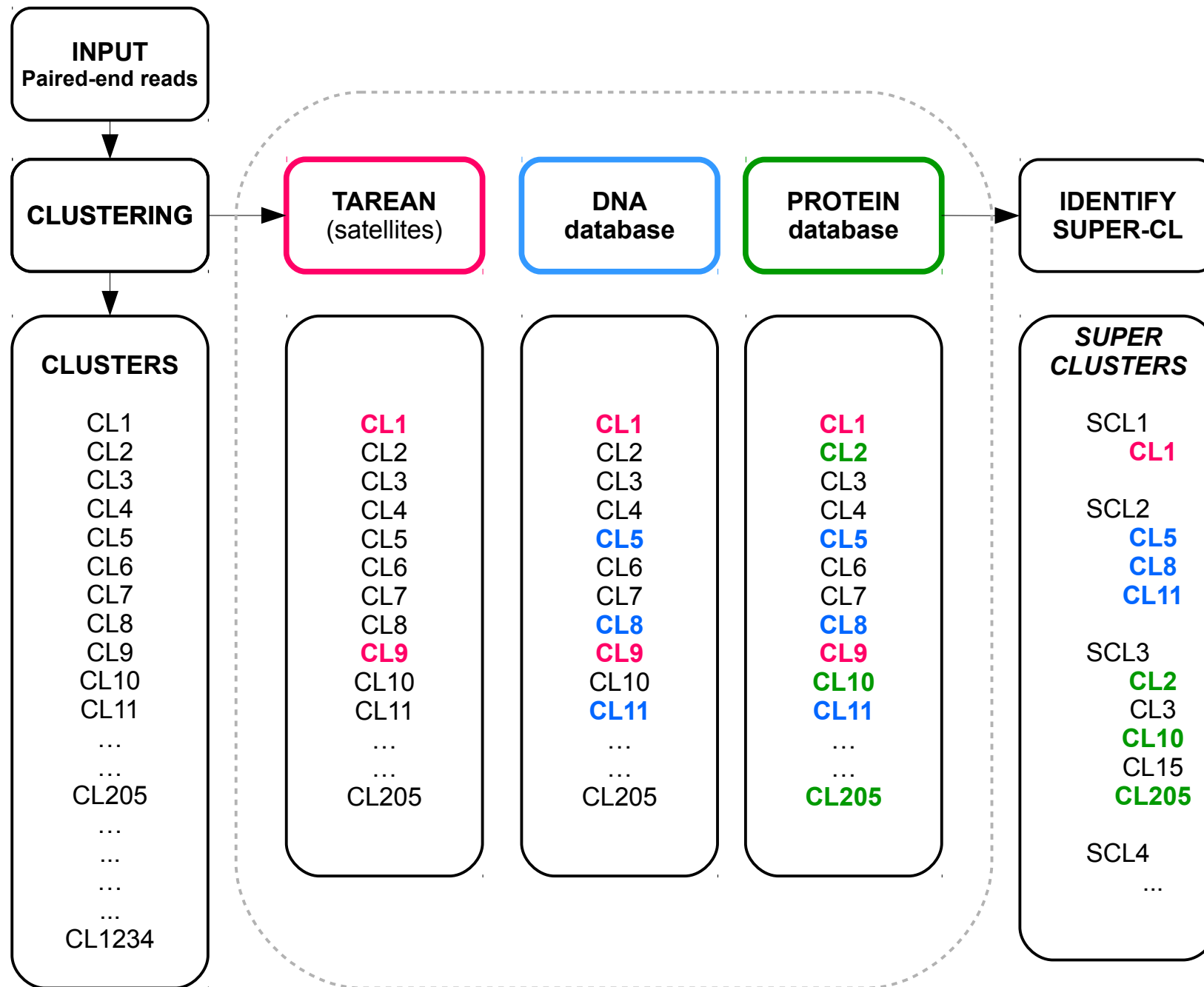


LTR-retrotransposon Ogre in *Vicia pannonica* (~ 40% of the genome)

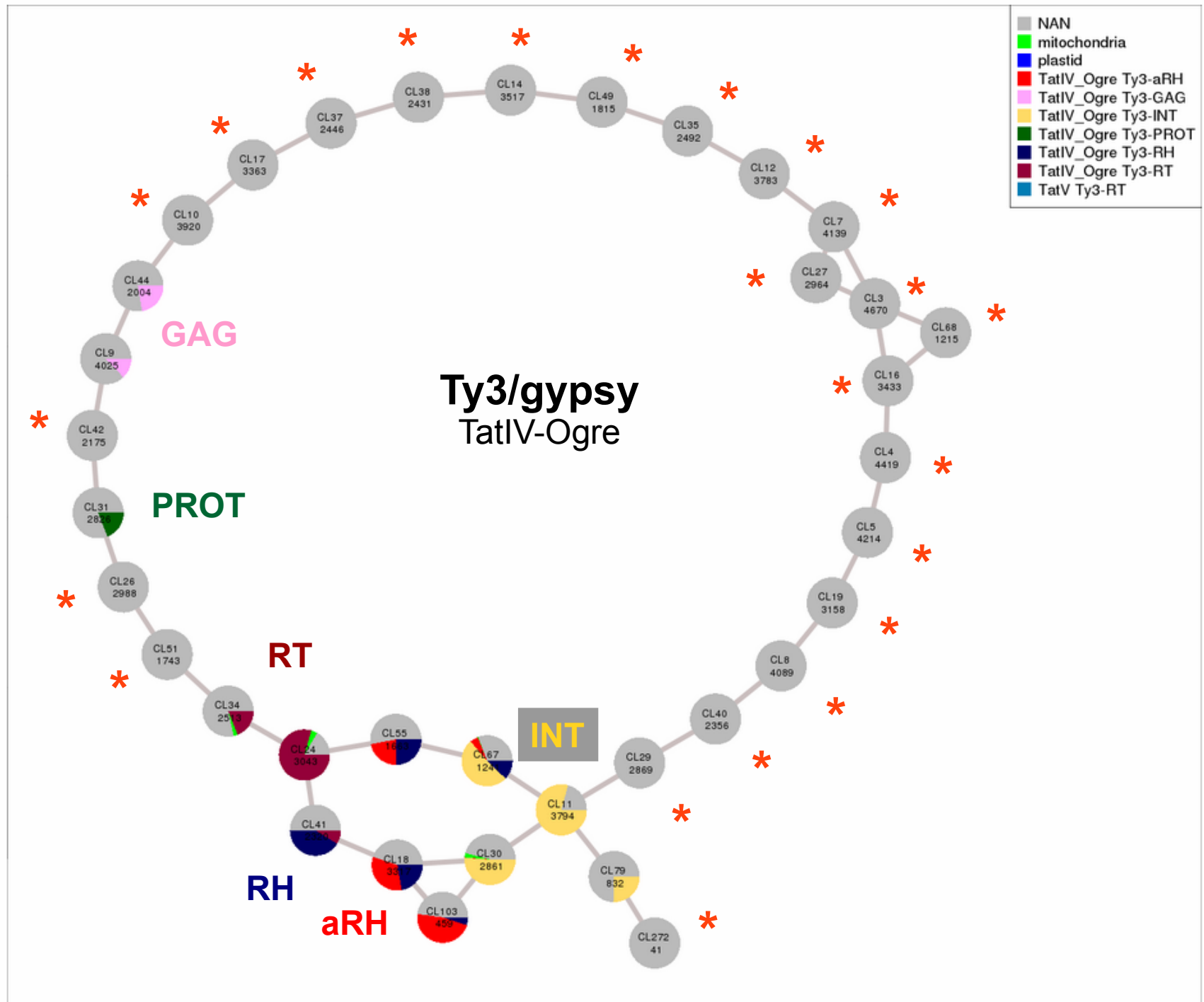
RepeatExplorer 2 – automatic detection of superclusters



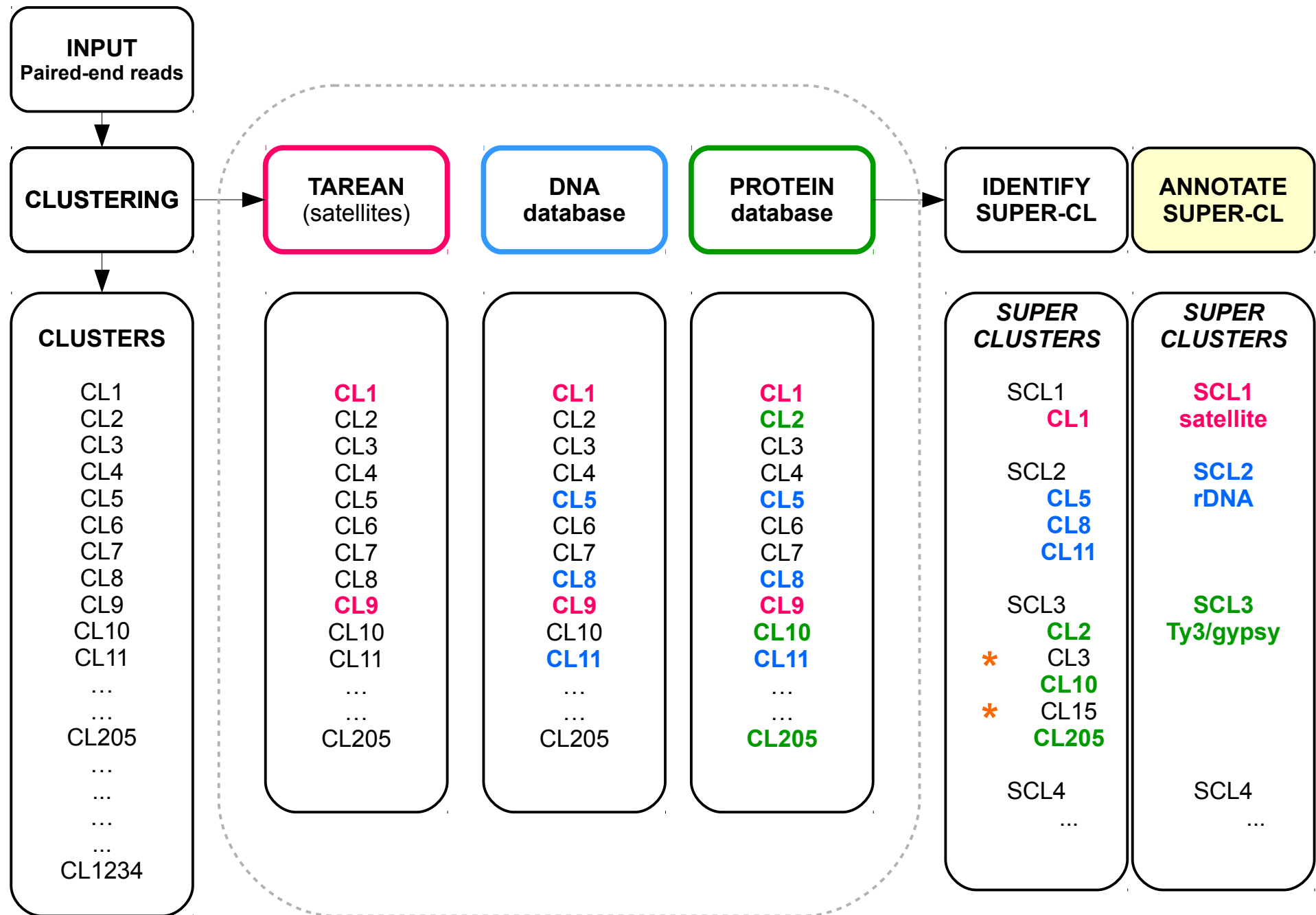
RepeatExplorer 2 – automatic detection of superclusters



Superclusters provide more complete annotation



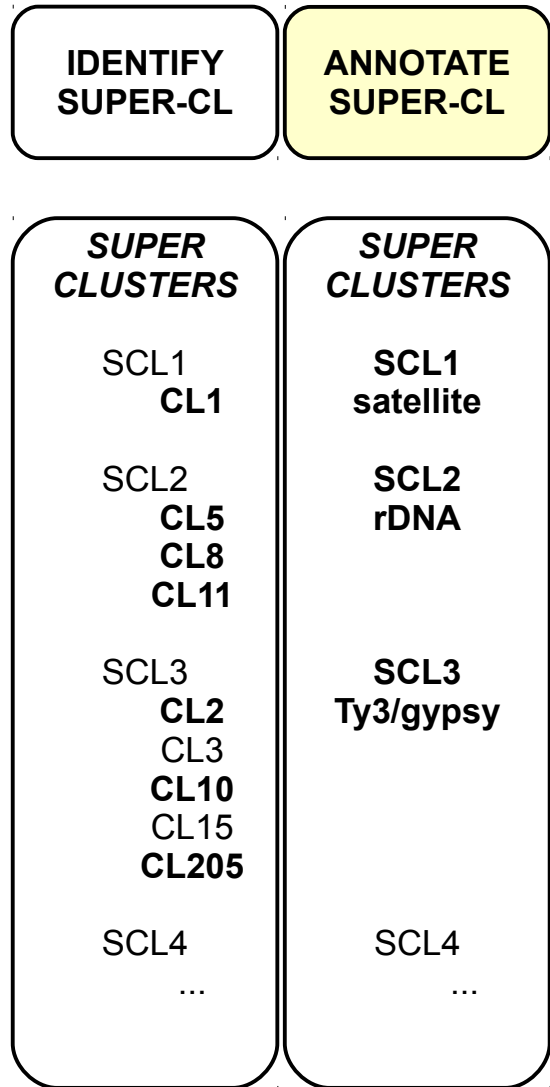
Superclusters provide more complete annotation



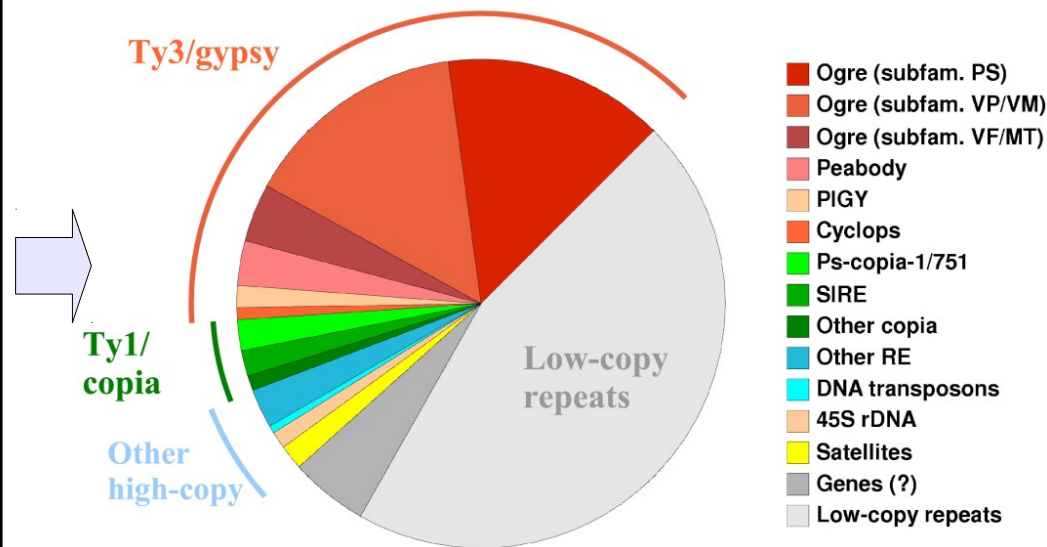
Repeat quantification

Cluster sizes (numbers of reads)
are proportional to genomic
abundance of corresponding repeats

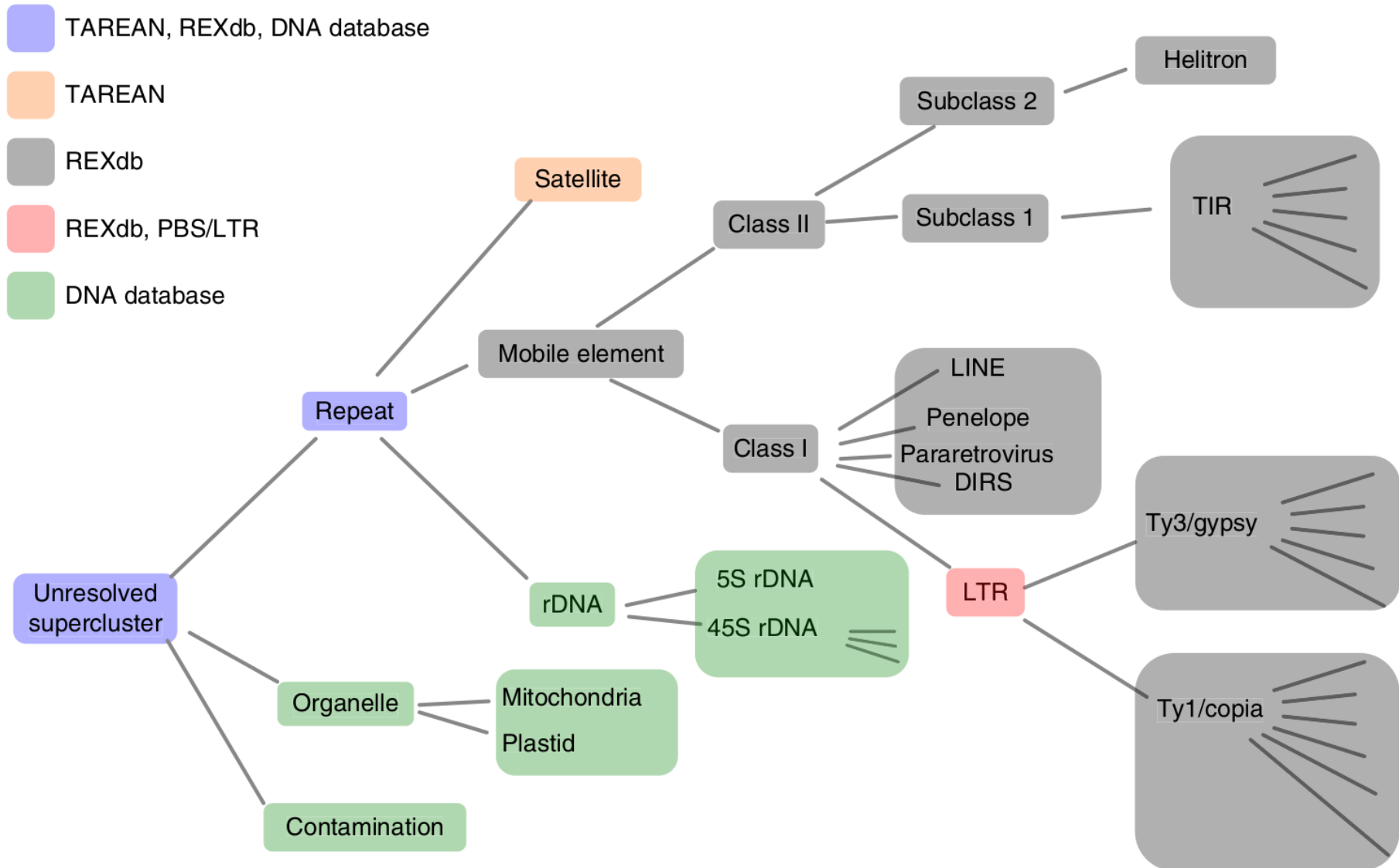
	7192836 (= 100%)	64.1
CL	reads	genome %
1	304159	4.229
2	234749	3.264
3	216307	3.007
4	202822	2.820
5	149693	2.081
6	145911	2.029
7	143766	1.999
8	142608	1.983
9	141836	1.972
10	123886	1.722
11	79345	1.103
12	72781	1.012
13	67096	0.933
14	65455	0.910
15	62334	0.867
16	53845	0.749
17	49341	0.686
18	45062	0.626
19	44762	0.622
20	43332	0.602
21	42344	0.589
22	40125	0.558
23	39923	0.555
24	36353	0.505
25	35977	0.500
26	35674	0.496
27	34829	0.484
28	34534	0.480
29	34302	0.477
30	33114	0.460
31	32930	0.458



Proportions of various repeat types in a genome

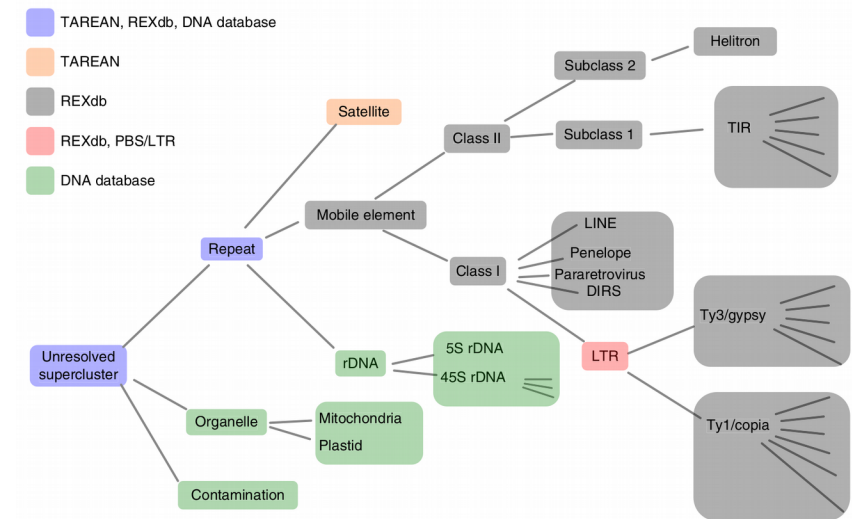


Decision tree for automatic annotation



How reliable is automatic annotation ?

- ✓ conserved repeats (rDNA)
- ✓ organelles, contamination
- ✓ satellite DNA (structure-based annotation)
- ✓ mobile elements (autonomous)
 - plants
 - other organisms
- ✗ non-autonomous mobile elements
 - ✓ LTR detection
 - insertion site detection (coming soon)



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- ✓ organelles, contamination
- ✓ satellite DNA (structure-based annotation)

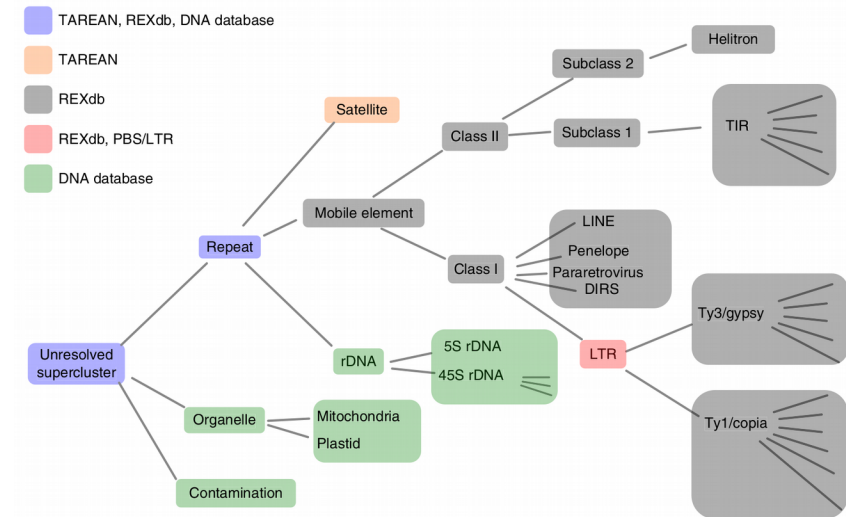
- ✓ mobile elements (autonomous)

- plants
- other organisms

- ✗ non-autonomous mobile elements

- ✓ LTR detection

- insertion site detection (coming soon)



Working with organisms not covered in REXdb ?

- use custom repeat database (and share it !)
- help us to improve REXdb ;-)

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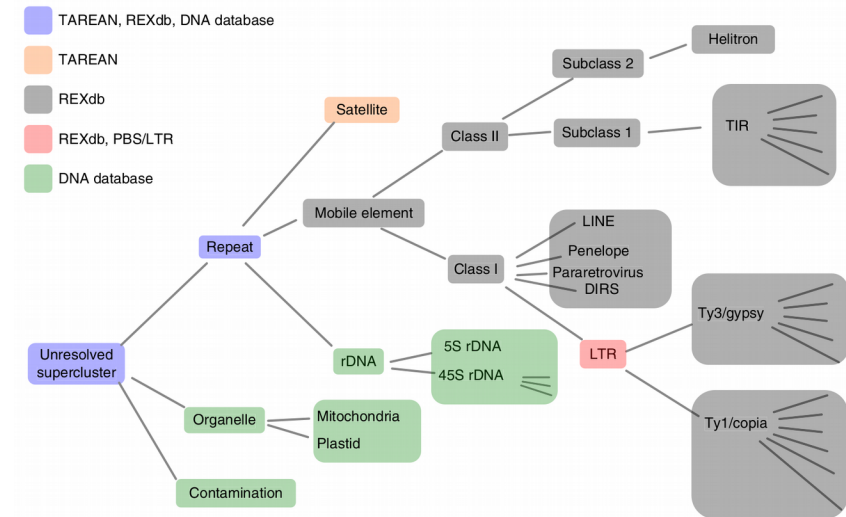
- ✓ mobile elements (autonomous)

- plants
- other organisms

- ✗ non-autonomous mobile elements

- ✓ LTR detection

- insertion site detection (coming soon)



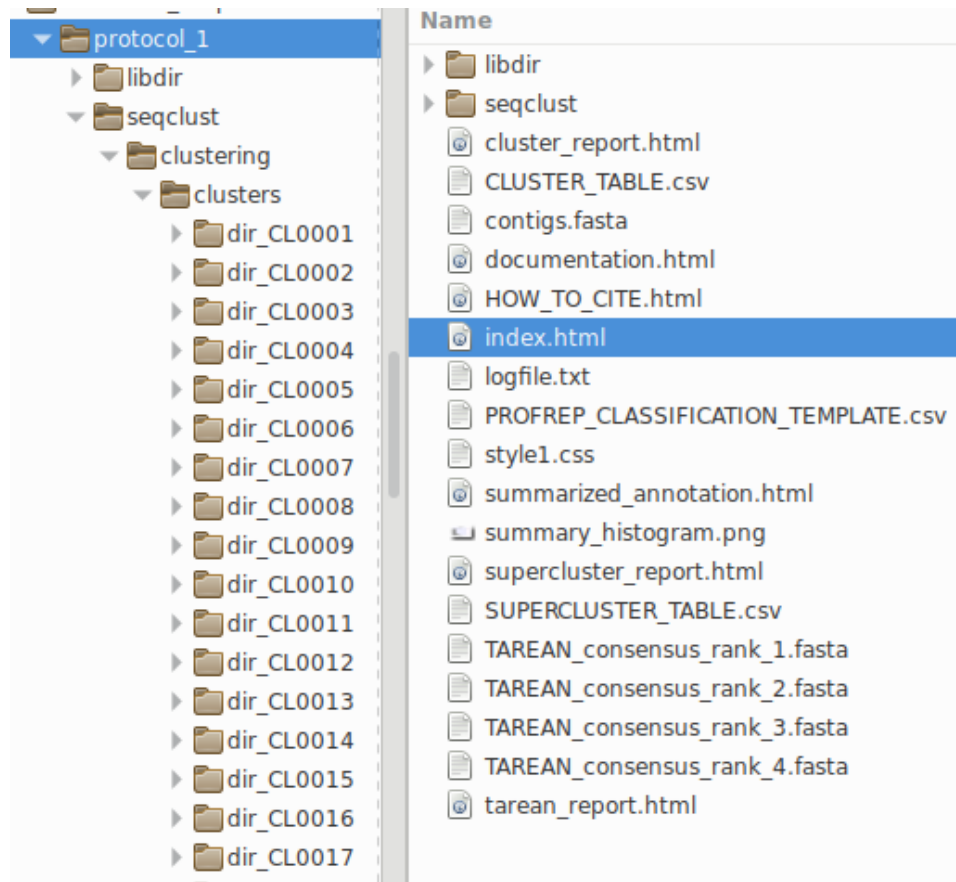
Working with organisms not covered in REXdb ?

- use custom repeat database (and share it !)
- help us to improve REXdb ;-)

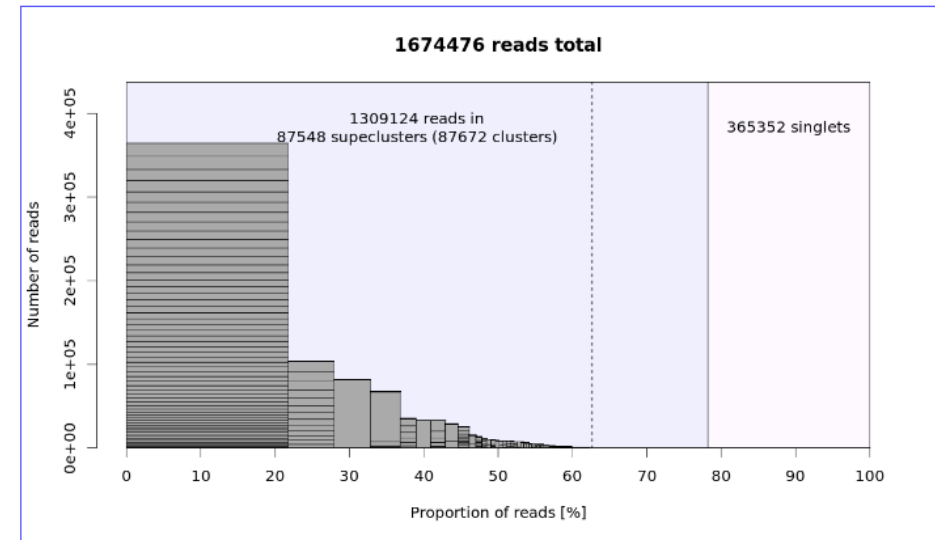
Clusters without annotation

- The lack of annotation does not mean these are not repeats (all repeats are detected)
- More info available in cluster directories

RepeatExplorer output files



Clustering Summary



Graphical summary of the clustering results. Bars represent superclusters, with their heights and widths corresponding to the numbers of reads in the superclusters (y-axis) and to their proportions in all analyzed reads (x-axis), respectively. Rectangles inside the supercluster bars represent individual clusters. If the filtering of abundant satellites was performed, the affected clusters are shown in green, and their sizes correspond to the adjusted values. Blue and pink background panels show proportions of reads that were clustered and remained single, respectively. Top clusters are on the left of the dotted line.

Run information:

Number of input reads: 2935772

Number of analyzed reads: 1674476

Proportion of reads in top clusters : 63 %

Cluster merging: No

Paired-end reads: Yes

Available analyses:

[Tandem repeat analysis](#)

[Cluster annotation](#)

[Supercluster annotation](#)

[Repeat annotation summary](#)

Always download and use archive with complete output of the analysis

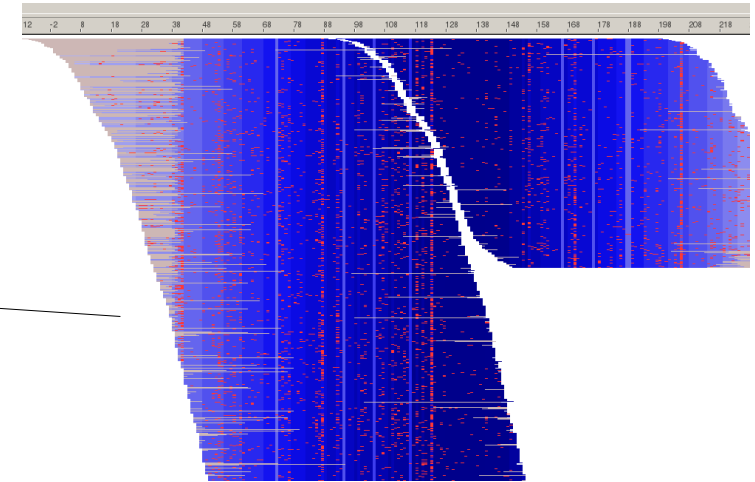
- do not work with HTML only

RepeatExplorer output files

Cluster directories

The screenshot shows the RepeatExplorer file explorer. On the left, a tree view shows the directory structure: Species_VVL > libdir > seqclust > clustering > clusters. The 'clusters' directory is expanded, showing a list of cluster directories from dir_CL0001 to dir_CL0025. The directory 'dir_CL0009' is selected and highlighted in blue. On the right, a list of files for 'dir_CL0009' is displayed under the 'Name' column. The files include: html_files, tarean, assembly.info, contigs.ace, contigs.aln, contigs.fasta, contigs.info.fasta, contigs.info.minRD5_sort-GR.fasta, contigs.info.minRD5_sort-length.fasta, contigs.info.minRD5_sort-RD.fasta, contigs.links, contigs.profile, contigs.qual, custom_db_extra_database_annotation.csv, dna_database_annotation.csv, graph_layout.GL, graph_layout.png, graph_layout_directed.RData, graph_layout_tmb.png, hitsort_part.csv, index.html, LTR_info.ADJ, LTR_info.LTR, LTR_info.with_PBS_blast.csv, protein_database_annotation.csv, reads.fasta, reads_selection.fasta, reads_selection_oriented.fasta, and singlets.fasta.

Name
html_files
tarean
assembly.info
contigs.ace
contigs.aln
contigs.fasta
contigs.info.fasta
contigs.info.minRD5_sort-GR.fasta
contigs.info.minRD5_sort-length.fasta
contigs.info.minRD5_sort-RD.fasta
contigs.links
contigs.profile
contigs.qual
custom_db_extra_database_annotation.csv
dna_database_annotation.csv
graph_layout.GL
graph_layout.png
graph_layout_directed.RData
graph_layout_tmb.png
hitsort_part.csv
index.html
LTR_info.ADJ
LTR_info.LTR
LTR_info.with_PBS_blast.csv
protein_database_annotation.csv
reads.fasta
reads_selection.fasta
reads_selection_oriented.fasta
singlets.fasta



assembly

Contigs

Cluster graph

LTR-detection

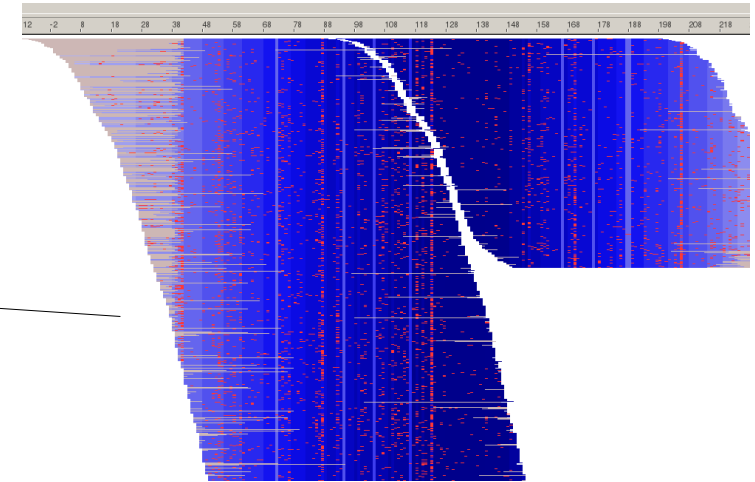
Read sequences

RepeatExplorer output files

Cluster directories

The screenshot shows the directory structure of RepeatExplorer output files. On the left, a tree view shows the path: Species_VVL > libdir > seqclust > clustering > clusters > dir_CL0009. The main panel displays a list of files for dir_CL0009. The files are:

Name
html_files
tarean
assembly.info
contigs.ace
contigs.aln
contigs.fasta
contigs.info.fasta
contigs.info.minRD5_sort-GR.fasta
contigs.info.minRD5_sort-length.fasta
contigs.info.minRD5_sort-RD.fasta
contigs.links
contigs.profile
contigs.qual
custom_db_extra_database_annotation.csv
dna_database_annotation.csv
graph_layout.GL
graph_layout.png
graph_layout_directed.RData
graph_layout_tmb.png
hitsort_part.csv
index.html
LTR_info.ADJ
LTR_info.LTR
LTR_info.with_PBS_blast.csv
protein_database_annotation.csv
reads.fasta
reads_selection.fasta
reads_selection_oriented.fasta
singlets.fasta



assembly

Contigs

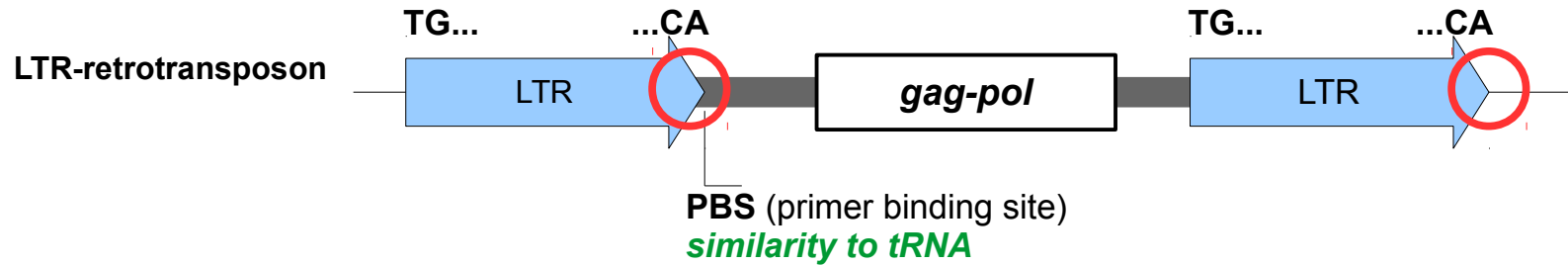
Cluster graph

LTR-detection

Read sequences

Insertion sites of mobile elements

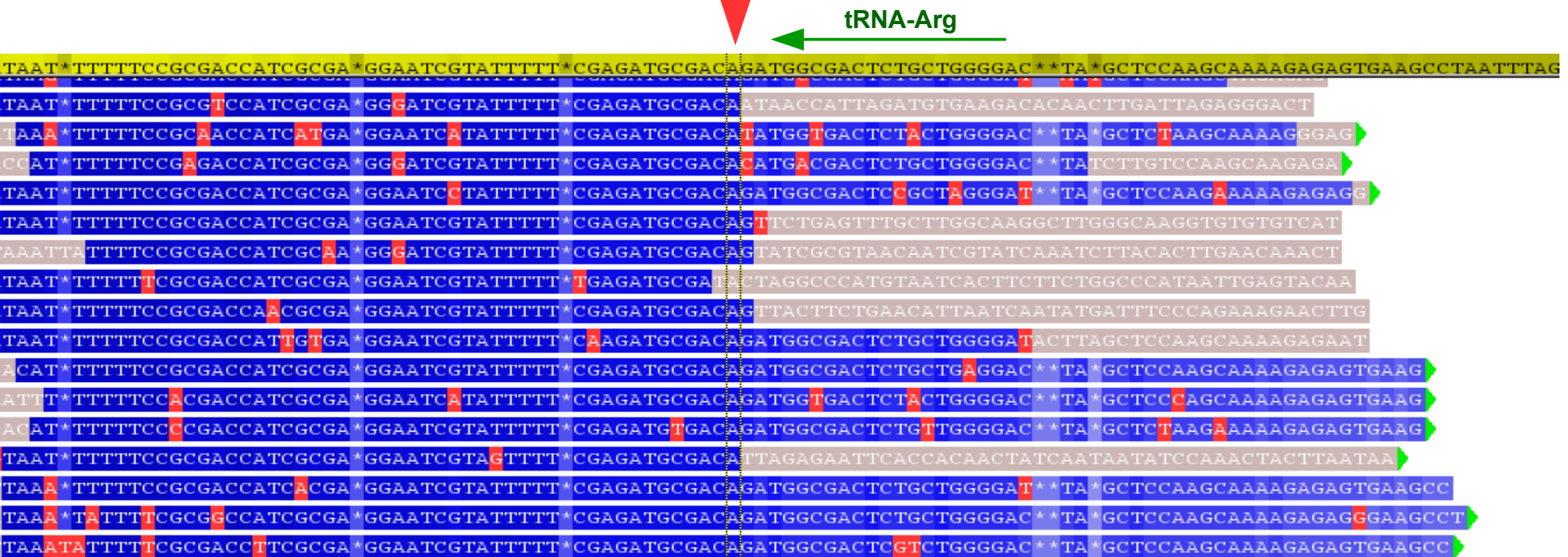
A tool for detection of LTR / PBS sites



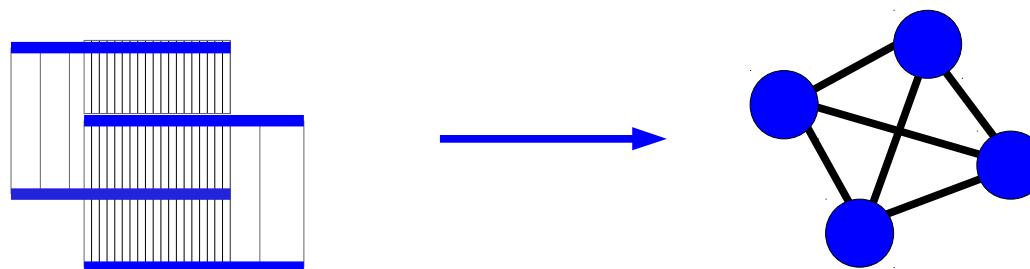
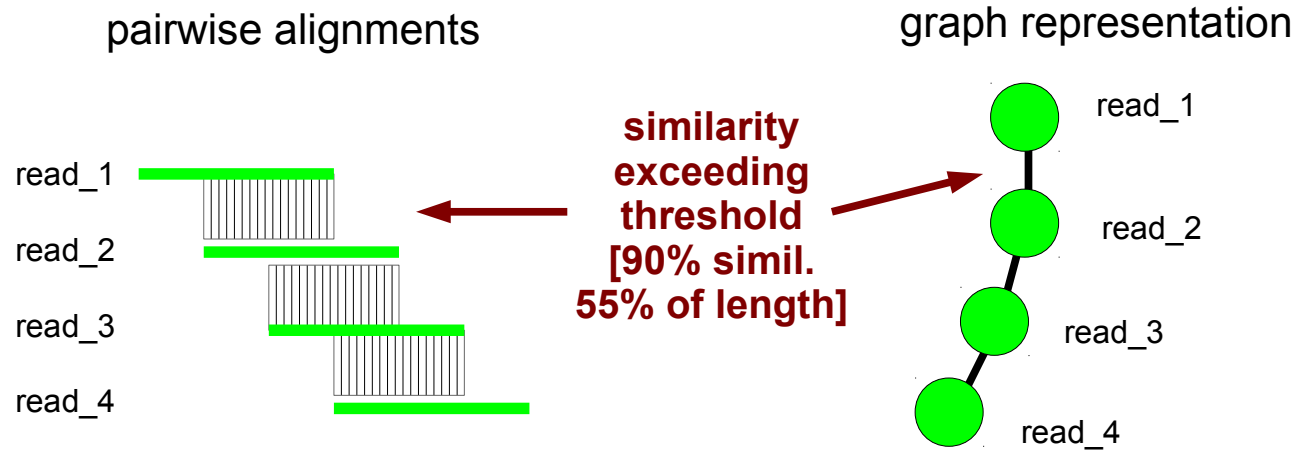
Program output:

CL	contig	pos.	site	site_depth	out_ma	maske	maske	region_in	region_out	blast to tRNA	%	length		site from	to	tRNA from	to	E-val
19	400	364	TGCGACA	106.6	30.4	0.0362	0.2975	GCGAGGAA	GATGGCGA	At-chr2.tRNA28-Arg	100	18	0	0	3	20	23	6 7E-007

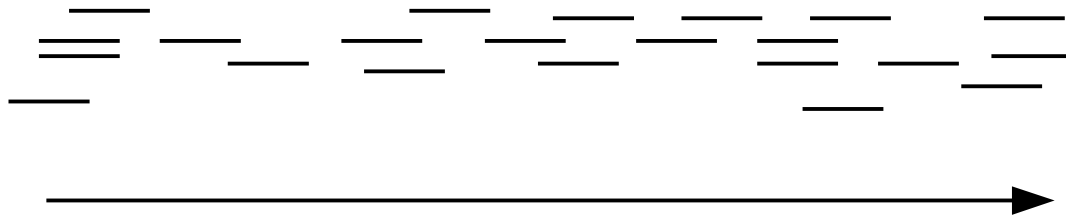
(window size 7)



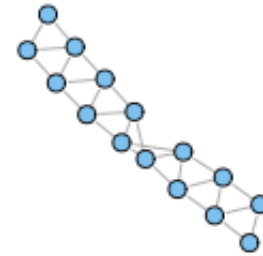
Graph shapes



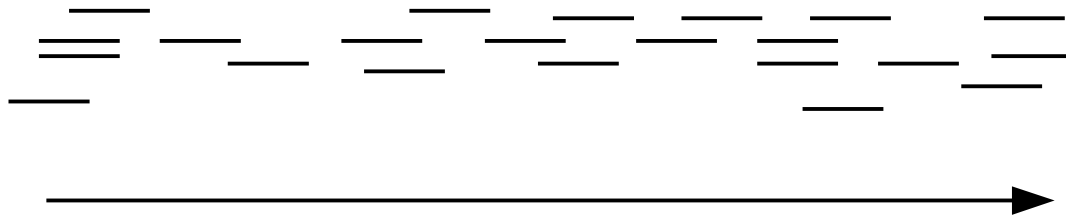
Linear graphs



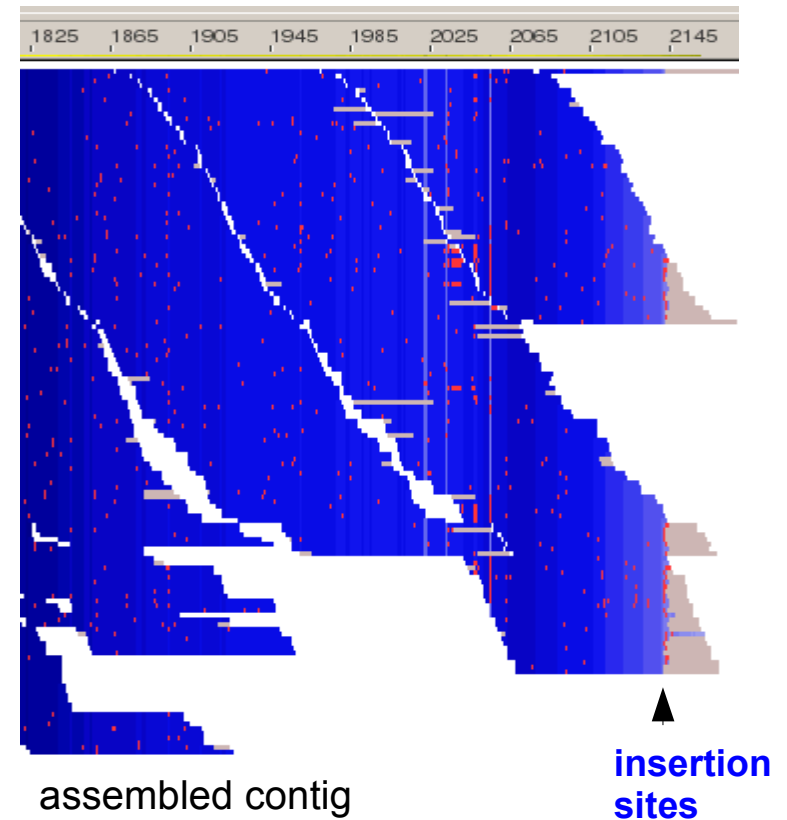
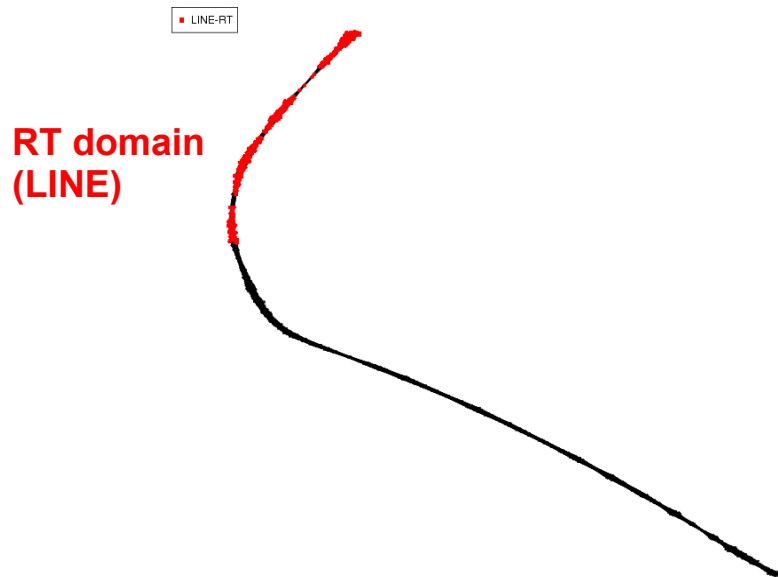
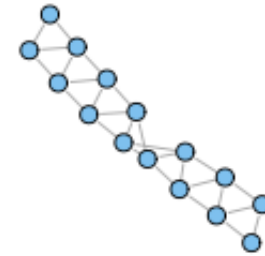
DNA transposons, LINEs, ...



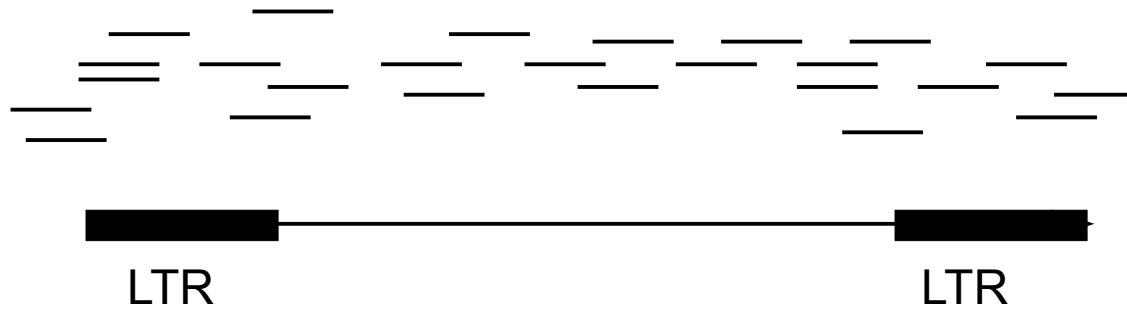
Linear graphs



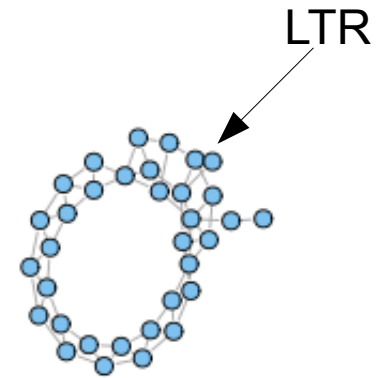
DNA transposons, LINEs, ...



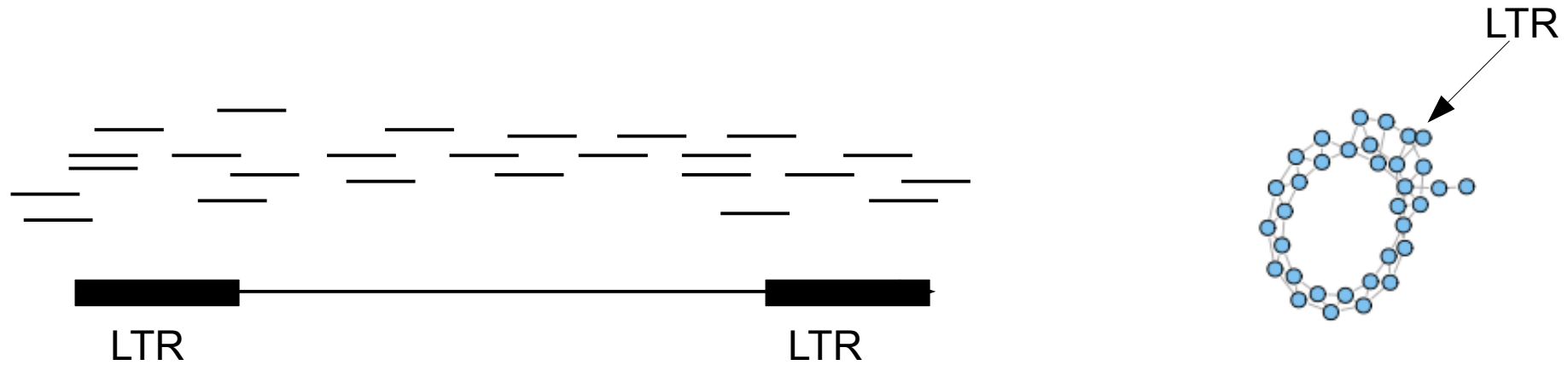
Linear / large circular graphs



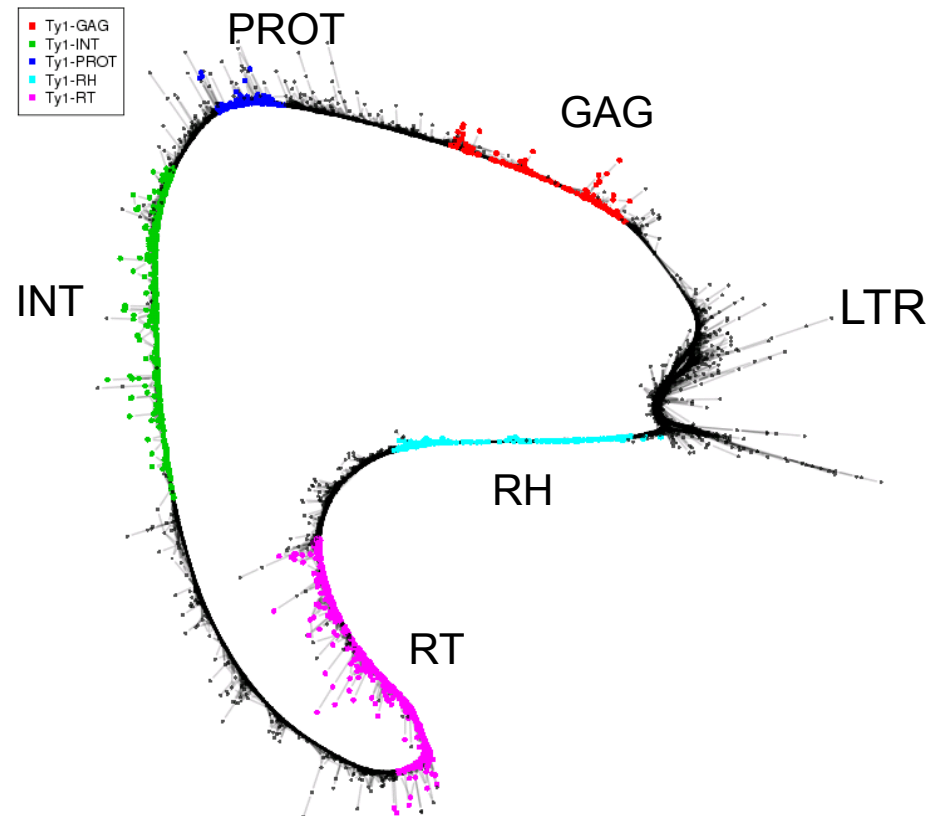
LTR-retrotransposons



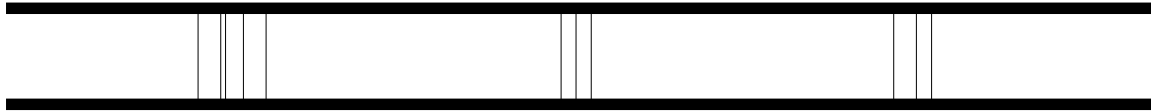
Linear / large circular graphs



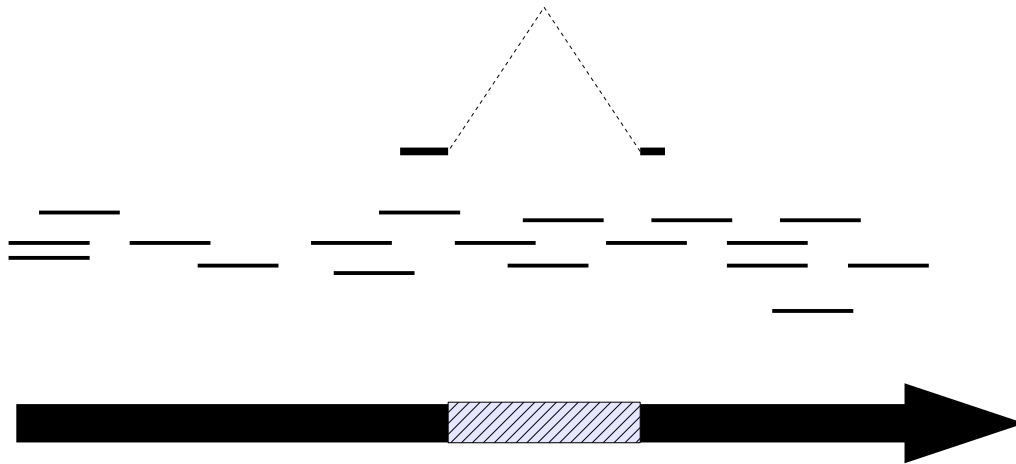
LTR-retrotransposons



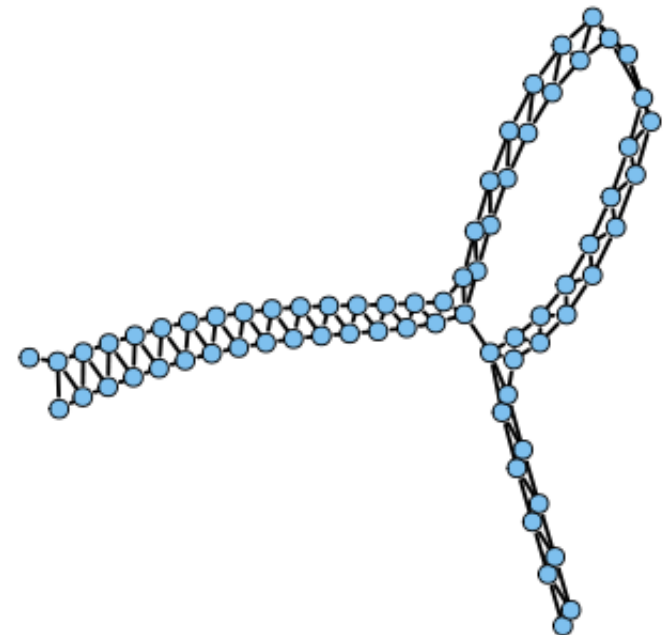
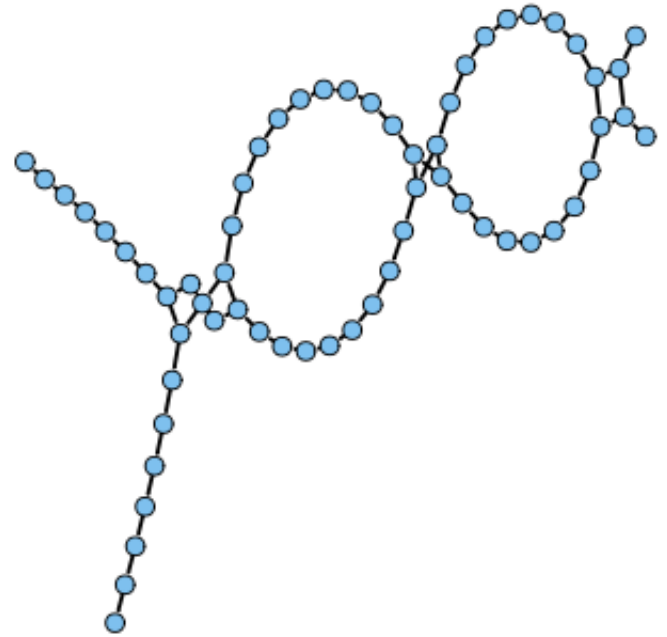
It's getting complicated...



Repeat variants with partial similarity
(depends on similarity threshold)



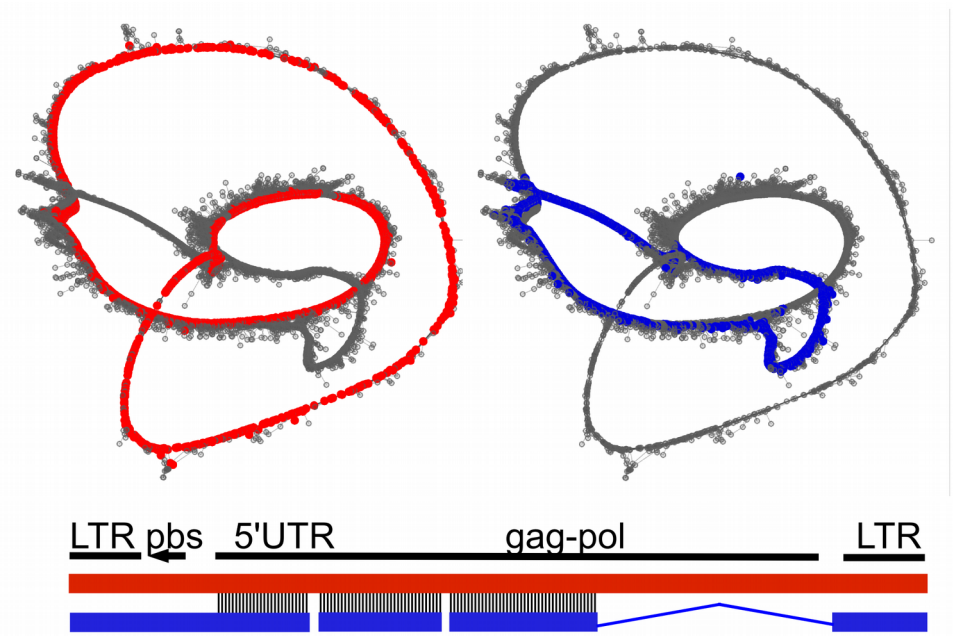
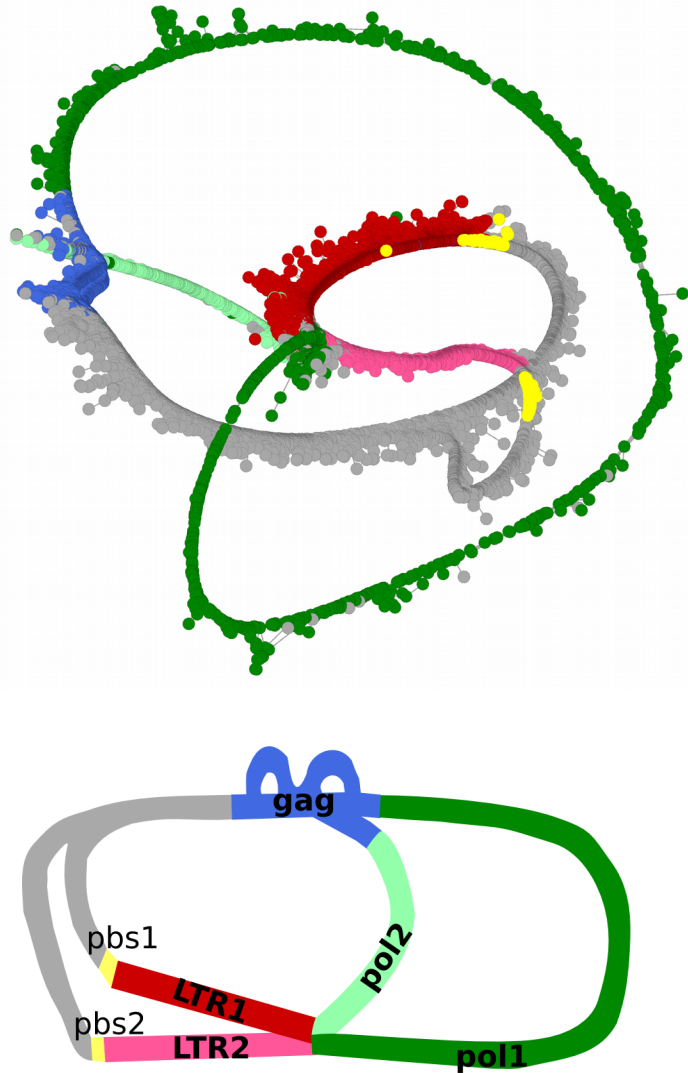
Variants differing in indel



It's getting complicated...

...but it has some meaning

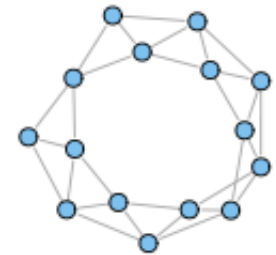
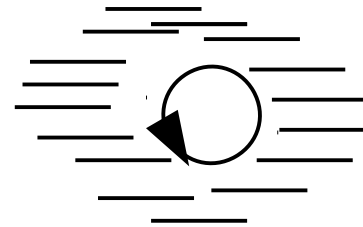
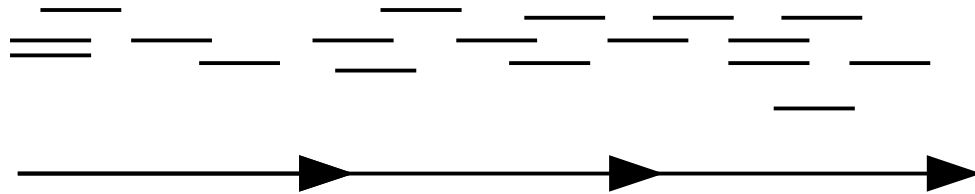
(in this case, presence of element variants differing in LTR sequence and in deletion within gag-pol region)



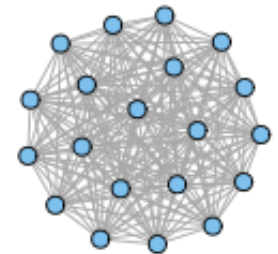
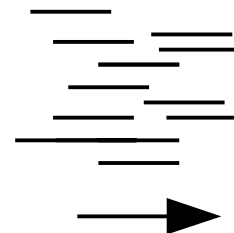
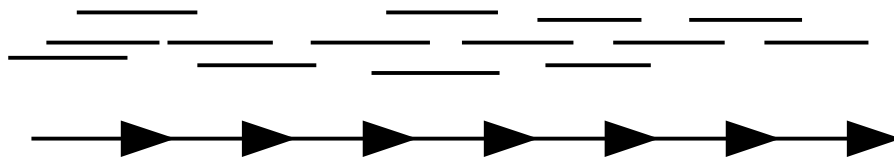
Circular graphs

TANDEM REPEATS

Read length $\ll$ monomer

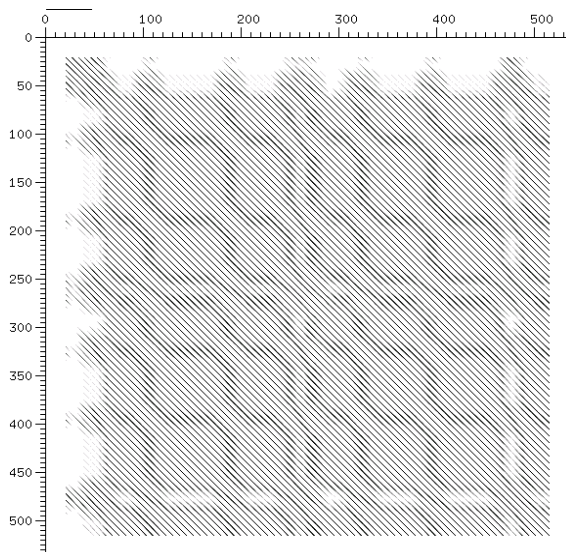
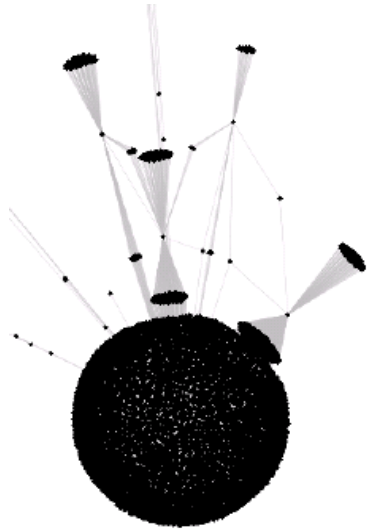


Read length $\geq$ monomer

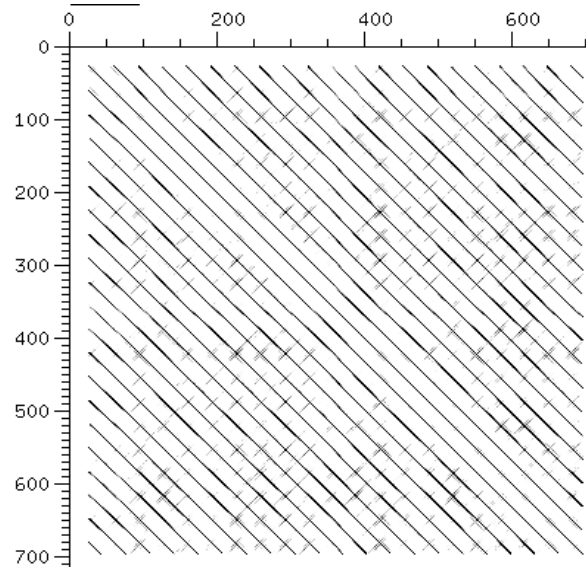
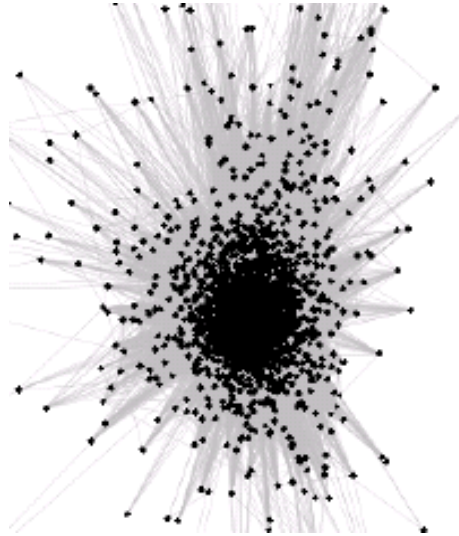


Circular graphs

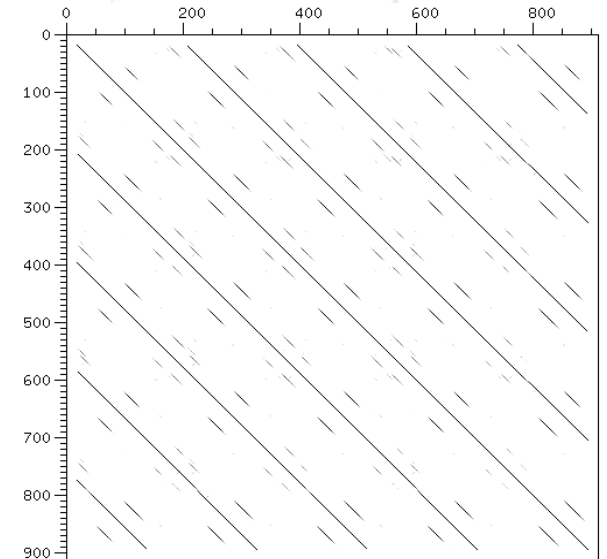
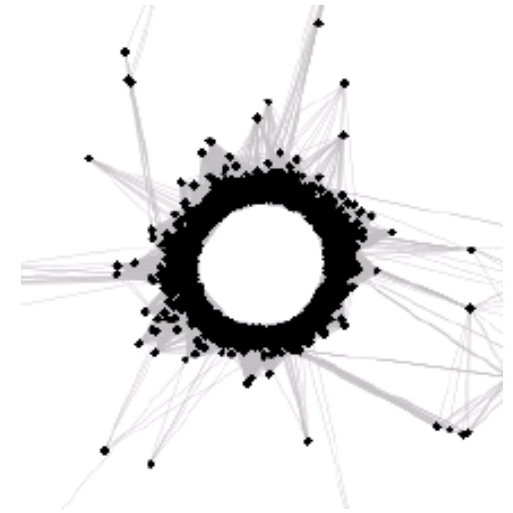
**microsatellite (6bp)
(GAACCT)_n**



satellite 35 bp

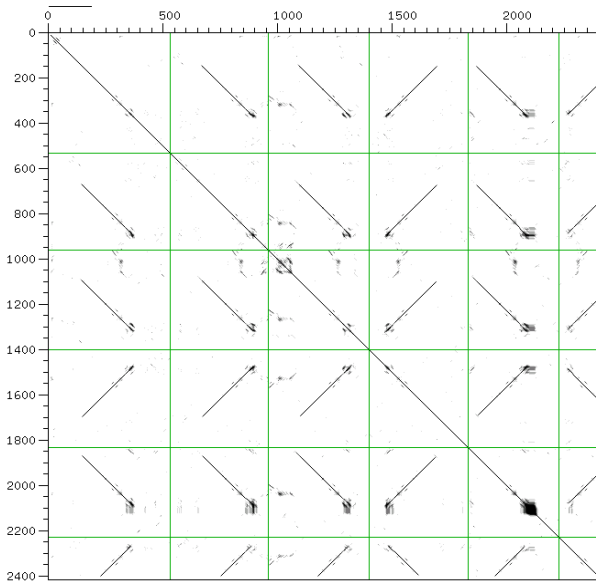
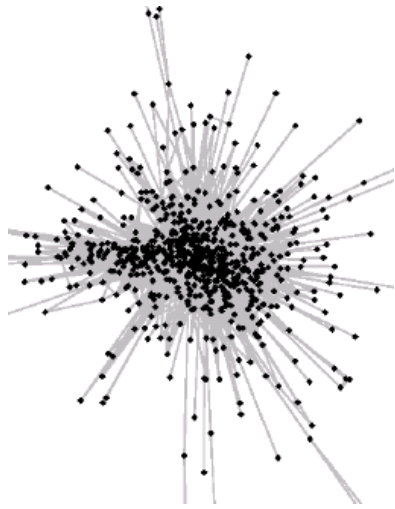


satellite 190 bp

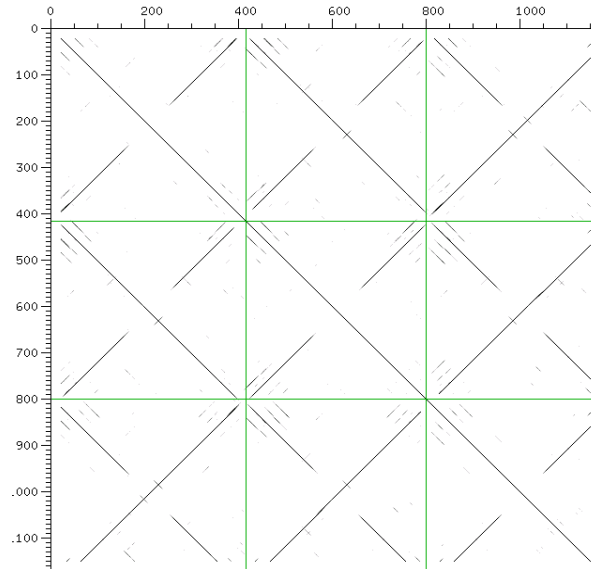


Circular graphs

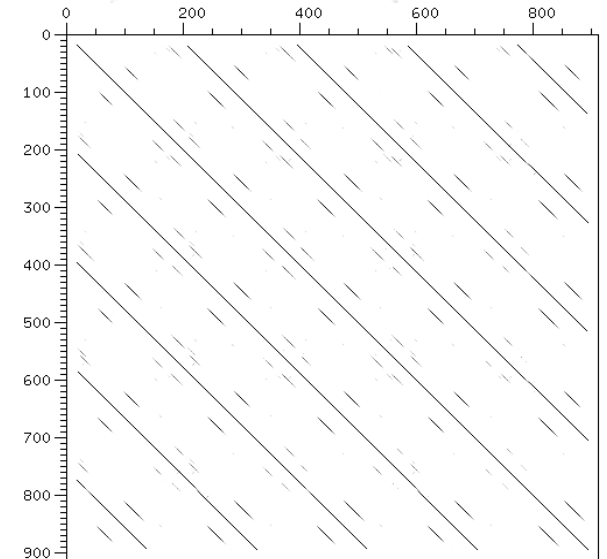
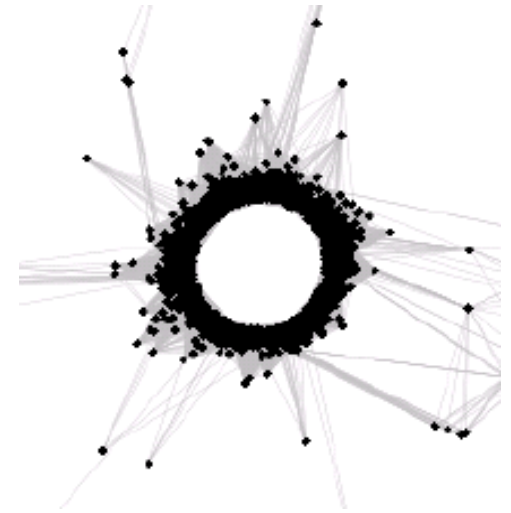
SINE



**MITE
(foldback)**

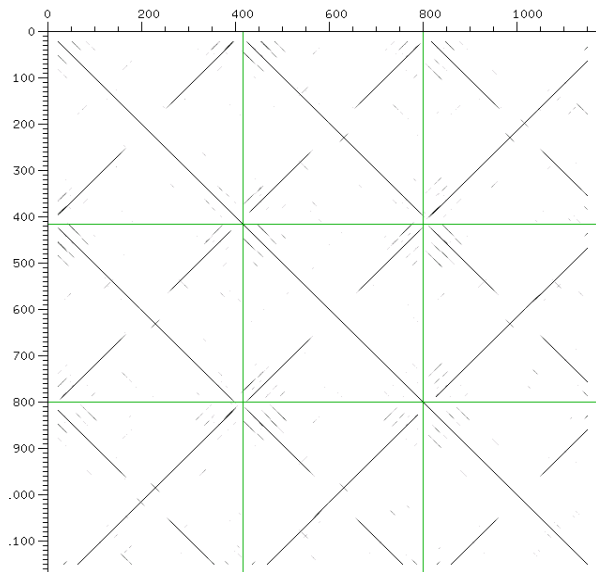


satellite 190 bp



Insertion sites of mobile elements

MITE (foldback)

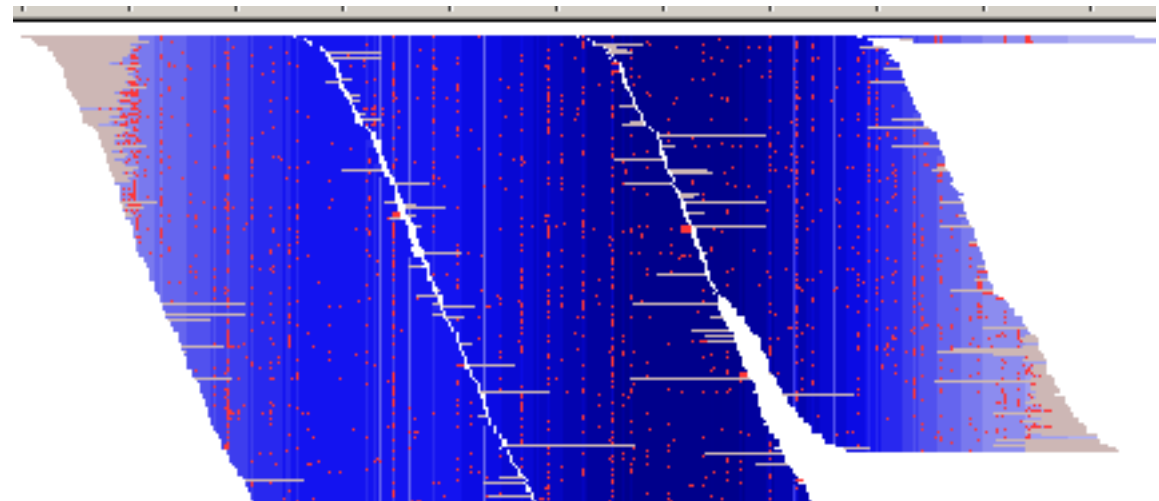


```
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CCTCGTAATAATAATTAATAAATTTTTGGATTTTCAACCATTTTAGTCATATT  
AAAATTGTATTTTAAATTTTTTTTAGGGTTTTCAACAATTTTAGTCATATT  
TTGAGCCAAACTAATAGTTAAATTTTAGGGTTTTCAACAATTTTAGTCATATT  
ACACATGCCATTTTATGATAGAAAAAGGGTTTTCAACAATTTTAGTCATATT  
TAGTAGTAGAAATGGTTTTTAGGGTTTTAGGGTTTTCAACAATTTTAGTCATATT  
TTTTATTATTTTGTGATTTTAAATAGGGTTTTGAACAATTTTAGTCATATT  
CTTAATTCAATTATAAAAGATTTAAAGGGTTTTTAACAATTTTAGTCATATT  
AAAATTGTATTTTAAATTTTTTTTAGGGTTTTCAACAATTTTAGTCATATT  
GCAGTTTATGGTATAAAATATTTAAAGGGTTTTCAATAATTTTAGTCATATT  
TGTAATTTTATTTGCATATTTTAAAGGGTTTTCAATATTTTGGTCATATT  
TTTGAAATAAAATTGAGTATATAAATAGGGTTTTCAACAATTTTATCATATT  
TTAATGGACTAAATGTGTATTTTAAAGGGTTTTCAACAATTTTAGTCATATT  
TTTTAACTAAATTGCATGTATTTTAGGGTTTTCAACAATTTTAGTCATATT  
TTCTATTTCAATCATGTATAAATTAAGGGTTTTCAACAATTTTAGTCATATT
```

← TIR

```
TGAAACAAATATGACCAAAAAAGTTAAAAACCCCTTAAATAAAACATAAAGAGTATAAA  
TGAAACAAATATGACCAAAAAAGTTAAAAACCCATTTTAAATTTAAACA  
TGAAACAAATATGACCAAAAAAGTTAAAAATCCATAGATTTAATGTGAAAA  
TAAACAAATATGACCAAAAAATTTGAAAAACCCATATATATATATGTGA  
TGAAACAAATATAACCAAAAAAGTTAAAAACCCCTTAAATAAAATATAGTTA  
TGAAACAAATATGACCAAAAAAGTTAAAAACCTATTTTTTTTTATAAGATTAT  
TGAAACAAATATAACCAAAAAAGTTAAAAACCCCTTAAATTATACCCATTTT  
TGAAACAAATATGACCAAAAAAGTTAAAAACCCATTTTAAATTTAAACAATTAT  
TGAAATAAATATGAACAAAAATTTTAAACCCCTATGAAATATTGTTATAAGGG  
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TGAAACAAATATGACCAAAAAAGTTAAAAACCCCTTAAATGAATTACAAATATGCGTG  
TAAACAAATATGACCAAAAAAGTTAAAAACCCAAATAAAACATAAAGAGTATAAA  
TAAACAAATATGACAAAAAAAGTGAAAAACCCAAATTGAAATCATACAAAAAGAAG  
TGAAATAAATATGACCAAAAAAGTTAAAAACCCGGTAATAAATAGAGTAAGCATATT  
TGAAACAAATATGACCAAAAAAGTTAAAAATCCATAGATTTAATGTGAAAAATACGAT  
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TGAAACAAATATAACCAAAAAAGTTAAAAACCCCTTAAATTATACCCATTTTGT
```

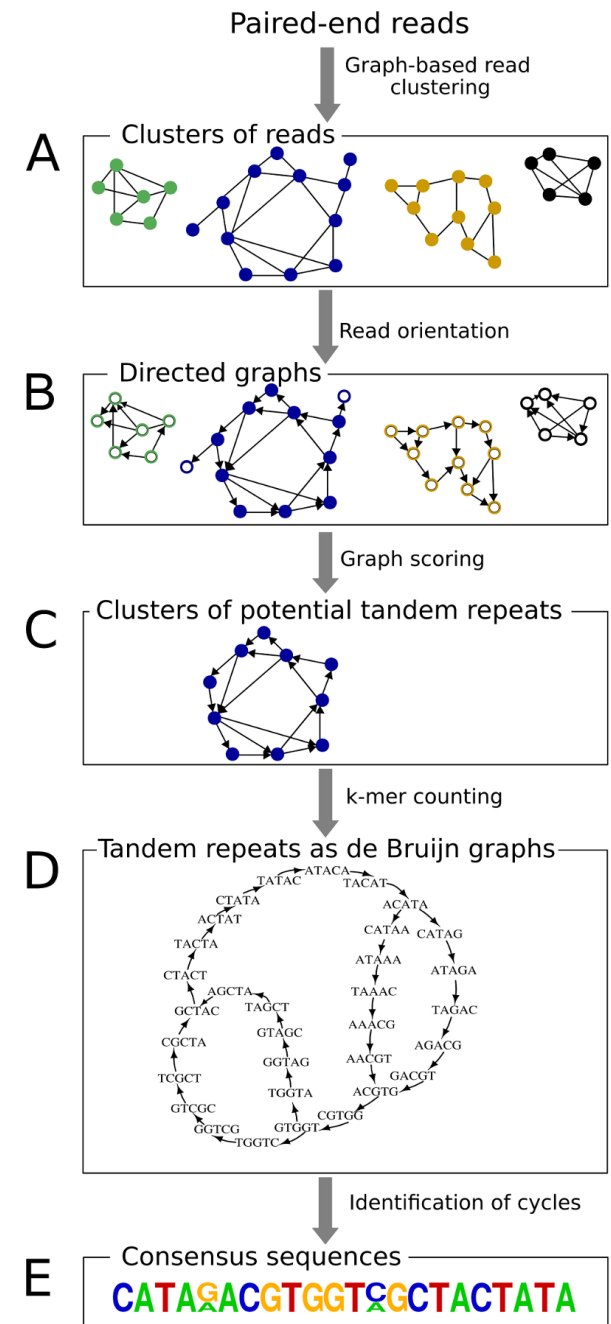
TIR →



contig

TAREAN

- Detects clusters with circular graphs automatically
- Calculates consensus sequences (alignment-free)
- Uses various parameters to distinguish tandem repeats from mobile elements
- *It is recommended to run TAREAN with cluster merging option as a complementary analysis to RepeatExplorer*



Using *RE* output for repeat annotation and quantification

- Check / correct automatic annotation
- Use all data available from the analysis archive (not just HTML output)
- Interpret the results considering your experimental setup
 - e.g., proportion of reads in clusters vs. singlet reads depends on coverage
 - efficiency of automatic annotation depends on organism (taxon, genome composition)
- *More info in our practical tutorials*